

Implement the Car Price Prediction Model using Python

CODE:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Mileage': [80000, 70000, 60000, 50000, 40000, 30000, 20000, 10000],
    'Engine': [1200, 1300, 1400, 1500, 1600, 1700, 1800, 2000],
    'Price': [400000, 450000, 500000, 600000, 700000, 800000, 950000,
1100000]
}

df = pd.DataFrame(data)
X = df[['Year', 'Mileage', 'Engine']]
y = df['Price']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Car Price Prediction Model")
print("-----")
```

```
print("\nActual Prices:")
print(y_test.values)

print("\nPredicted Prices:")
print(y_pred)

print("\nMean Squared Error:", mse)
print("R2 Score:", r2)

# Predict price for a new car
new_car = [[2021, 25000, 1500]]
predicted_price = model.predict(new_car)

print("\nNew Car Details:")
print("Year: 2021")
print("Mileage: 25000 km")
print("Engine: 1500 cc")

print("\nPredicted Car Price: Rs.", round(predicted_price[0], 2))
```

OUTPUT:

... Car Price Prediction Model

Actual Prices:

[450000 800000]

Predicted Prices:

[445000. 815000.]

Mean Squared Error: 125000000.0000035

R2 Score: 0.9959183673469386

New Car Details:

Year: 2021

Mileage: 25000 km

Engine: 1500 cc

Predicted Car Price: Rs. 611250.0